# The orbit budget of a small singleton prefix

Human Compression project: L044 continuation

Retain the complete critical singleton hypotheses and prefix-growth proof. In particular $G \leq A_n$ is transitive with connected string C-marking, $G_0$ is the actual natural alternating group off $a$, and all maximal ambient omissions are intransitive. For $4 \leq g < r$ put

$$P = G_{<g}, \quad Y = \mathrm{supp}(P), \quad C = a^P, \quad H = G_{\{2,\dots,g-1\}}, \quad b = a^{\rho_0}.$$

Assume $P$ is intransitive on $Y$, $|Y| \leq 2g+1$, and each colour $2, \dots, g-1$ has at least two ambient-eligible edges. These are the actual hypotheses at the first shared index of the least later unique cut. All eligibility below refers to the full ambient omission, never to an orbit projection.

**Lemma 1.** *Suppose $\rho_0 = (a\ b)(u\ v)$, the set $E = \{u, v\}$ is preserved by $\rho_1$ and $\rho_2$, and $\rho_1$ fixes $a$ but moves $b$ to a point $x \notin \{a, b, u, v\}$. Then $G$ is intransitive.*

*Proof.* The $\langle \rho_0, \rho_1 \rangle$-orbit of $a$ is $\{a, b, x\}$. The set $E$ is an orbit of size two. Every other nontrivial orbit has size two with $\rho_0$ acting trivially, since $\rho_0$ fixes every point outside $\{a, b, u, v\}$. Any permutation centralizing both marks maps their orbits equivariantly. It must preserve $E$, the only two-point orbit on which $\rho_0$ acts nontrivially. Every mark of index at least three centralizes both marks; the first three preserve $E$ by hypothesis. Thus $G$ preserves the proper nonempty set $E$. □

**Theorem 2.** *Let $f = |Y \cap \mathrm{Fix}(H)|$, and let $c$ count the nontrivial $H$-orbits. Then $c \geq 2$, and precisely the following numerical possibilities remain:*

| $f$ | $c$ | $\lvert Y \rvert$ |
|---|---|---|
| 2 | 2 | $2g$ *or* $2g+1$ |
| 3 | 2 | $2g+1$ |
| 2 | 3 | $2g+1$ |

(1)

*Moreover $\rho_0$ cannot exchange the points of any two-point $H$-orbit. On the $H$-fixed points in $Y$, its action is exactly $(a\ b)$. Its odd action on the remaining points consequently has one of the following contributions: a free involution on a preserved $H$-orbit of size $2q$, where $q \geq 3$ is odd; or an exchange of two $H$-orbits of the same odd size at least three. This is a reduction to those possibilities, not their exclusion.*

*Proof.* The group $H$ fixes $a, b$: all its marks fix $a$ and commute with $\rho_0$. Both points lie in $Y$, so $f \geq 2$. The full ambient-interval theorem applies to $H$ and the given eligible pairs. Every moved point of $H$ lies in $Y$, hence

$$2(g-2) \leq |Y| - f - c, \qquad f + c \leq 5. \tag{2}$$

Also $c \geq 1$ because the interval is nonempty and its marks are nonidentity. Strict prefix growth gives $|C| \geq g+1$.

Suppose first that $c = 1$, and denote the sole nontrivial $H$-orbit by $U$. Then $f \leq 4$. Since $|C| \geq 5 > f$, the orbit $C$ meets $U$, hence contains it. A nontrivial $P$-orbit outside $C$ must lie among the $H$-fixed points. Such an orbit has at least two points, and $a, b$ already lie in $C$. Thus $f = 4$ and the other $P$-orbit is $E = \{u, v\}$; the fixed set in $Y$ is exactly $\{a, b, u, v\}$. Inequality (2) now forces $|Y| = 2g+1$ and $|U| = 2g-3$.

The mark $\rho_0$ centralizes $H$ and preserves its unique nontrivial orbit $U$. A centralizing involution on an odd transitive orbit fixes that orbit pointwise: it has a fixed point, and commutation then propagates fixation to the whole orbit. On the four remaining points $\rho_0$ swaps $a, b$, so its evenness forces $\rho_0 = (a\ b)(u\ v)$. The mark $\rho_1$ preserves $E$ because $E$ is a $P$-orbit. Strict growth makes it move $b$; its image is in $U$, since it fixes $a$ and preserves $E$. Finally $\rho_2 \in H$ fixes $E$ pointwise. Lemma 1 contradicts ambient transitivity. Thus $c \geq 2$.

Suppose next that $E = \{u, v\}$ is an $H$-orbit on which $\rho_0$ interchanges the two points. No edge of any colour $2, \ldots, g-1$ on $E$ is ambient-eligible: its endpoints are already joined by $\rho_0$ in the corresponding omission. All eligible edges of that interval are therefore confined to the union $V$ of the other nontrivial $H$-orbits. The complete confinement lemma, including its actual orbit count, gives

$$2(g-2) \leq |V| - (c-1) = |Y| - f - c - 1, \qquad f + c \leq 4. \tag{3}$$

There is at least one such other orbit, since every colour has eligible edges. Since $f \geq 2, c \geq 2$, equality forces $f = c = 2$, $|Y| = 2g + 1$, and $V$ is one orbit of odd size $2g - 3 \geq 5$. The mark $\rho_0$ preserves $V$, because its size differs from two, and fixes it pointwise by the same centralizer argument. Thus again $\rho_0 = (a\ b)(u\ v)$. The orbit $C$ has at least five points, whereas $Y \setminus V$ has only four, so $V \subseteq C$. If $C$ met $E$ it would contain all of $Y$, contrary to intransitivity of $P$ on $Y$. Consequently $E$ is the other $P$-orbit, preserved by $\rho_1$ and $\rho_2$. Strict growth makes $b^{\rho_1}$ lie in $V$. Lemma 1 gives the contradiction again.

The table follows immediately from (2), $f \geq 2, c \geq 2$ and $|Y| \leq 2g + 1$. Centralization of $H$ makes $\rho_0$ preserve its fixed set in $Y$. Its size is two or three, and $\rho_0$ already swaps $a, b$, so it fixes any third point. It therefore acts oddly on the nontrivial $H$-orbits. On a preserved transitive orbit, a nonidentity centralizing involution is free; its sign is odd exactly when the orbit has size $2q$ with $q$ odd. The two-point case was just excluded, so $q \geq 3$. An exchange of two orbits contributes the sign $(-1)^m$, where $m$ is their common size. An odd contribution of that form thus has odd $m \geq 3$. These possibilities exhaust the odd sign and prove the asserted reduction. □

**Attribution and boundary.** The complete ambient-interval bound, confinement lemma and actual singleton prefix growth are acquired inputs. The application to an initial interval, its small fixed-set budget, and the uniform invariant-pair exclusions are the present continuation. The source setting is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for $A_n$*, Propositions 5.28–5.32. The two- and three-orbit cases in (1) have not been eliminated. No faithful projected marking, projected IP, or pure alternating prefix kernel is assumed. There is no new finite computation, formal declaration, whole-original recovery or question credit.